IE307 – Công nghệ lập trình đa nền tảng cho ứng dụng di động

BÀI TẬP THỰC HÀNH 1: COMPONENTS, LAYOUT, PROPS & STATE

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Lớp: IE307.R11.1

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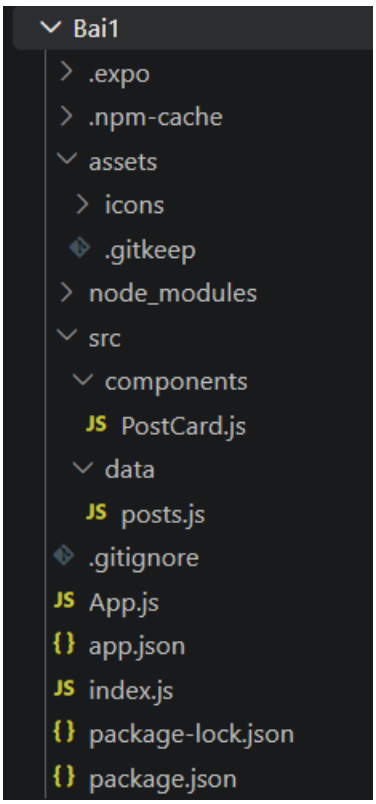
Bài làm

Bài 1: Social Media Feed

Link video

https://drive.google.com/file/d/1XwjH9pO4JqF1ZBwReYJFty9pe9GcS8MC/view?usp=drive_link

Cấu trúc project



Các file code

App.js

Bai1 > JS App.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { useState } from 'react';
import {
  FlatList,
  Platform,
  SafeAreaView,
  StatusBar as NativeStatusBar,
  StyleSheet,
  Text,
  View,
} from 'react-native';
import { StatusBar } from 'expo-status-bar';

import PostCard from './src/components/PostCard';
import initialPosts from './src/data/posts';

export default function App() {
  const [posts, setPosts] = useState(initialPosts);

  const updatePost = (postId, updatePostData) => {
    setPosts((currentPosts) =>
      currentPosts.map((post) => (post.id === postId ? updatePostData(post) : post)),
    );
  };

  const handleLike = (postId) => {
    updatePost(postId, (post) => ({
      ...post,
      liked: !post.liked,
      likes: Math.max(0, post.likes + (post.liked ? -1 : 1)),
    }));
  };

  const handleComment = (postId) => {
    updatePost(postId, (post) => ({
      ...post,
      comments: post.comments + 1,
    }));
  };

  const handleShare = (postId) => {
    updatePost(postId, (post) => ({
      ...post,
      shares: post.shares + 1,
    }));
  };

  const renderPost = ({ item }) => (
    <PostCard
      post={item}
      onLike={() => handleLike(item.id)}
      onComment={() => handleComment(item.id)}
      onShare={() => handleShare(item.id)}
    />
```

```
  );

  return (
    <SafeAreaView style={styles.safeArea}>
      <StatusBar style="light" />
      <View style={styles.header}>
        <Text style={styles.headerTitle}>Social Media Feed</Text>
        <Text style={styles.headerSubtitle}>Connect, share, and enjoy</Text>
      </View>
      <FlatList
        data={posts}
        renderItem={renderPost}
        keyExtractor={(item) => item.id}
        contentContainerStyle={styles.feedContent}
        showsVerticalScrollIndicator={false}
      />
    </SafeAreaView>
  );
}

const styles = StyleSheet.create({
  safeArea: {
    flex: 1,
    backgroundColor: '#eef2f8',
  },
  header: {
    paddingHorizontal: 20,
    paddingTop: Platform.OS === 'android' ? NativeStatusBar.currentHeight + 13 : 13,
    paddingBottom: 15,
    backgroundColor: '#1264d7',
  },
  headerTitle: {
    color: '#ffffff',
    fontSize: 24,
    fontWeight: '800',
  },
  headerSubtitle: {
    marginTop: 3,
    color: '#dce9ff',
    fontSize: 13,
  },
  feedContent: {
    padding: 12,
    paddingBottom: 28,
  },
});
```

PostCard.js

Bai1 > src > components > JS PostCard.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { Image, Pressable, StyleSheet, Text, View } from 'react-native';

const actionIcons = {
  like: require('../../assets/icons/like.png'),
  liked: require('../../assets/icons/like-filled.png'),
  comment: require('../../assets/icons/comment.png'),
  share: require('../../assets/icons/share.png'),
};

export default function PostCard({ post, onLike, onComment, onShare }) {
  return (
    <View style={styles.card}>
      <View style={styles.authorRow}>
        <Image source={{ uri: post.avatar }} style={styles.avatar} />
        <View style={styles.authorText}>
          <Text style={styles.username}>{post.username}</Text>
          <Text style={styles.postTime}>Just now</Text>
        </View>
      </View>

      <Text style={styles.content}>{post.content}</Text>
      <View style={styles.postImageFrame}>
        <Image source={{ uri: post.image }} style={styles.postImage} />
      </View>

      <View style={styles.statisticsRow}>
        <Text style={styles.statisticsText}>{post.likes} Likes</Text>
        <Text style={styles.statisticsText}>{post.comments} Comments</Text>
        <Text style={styles.statisticsText}>{post.shares} Shares</Text>
      </View>

      <View style={styles.divider} />

      <View style={styles.actionsRow}>
        <Pressable
          accessibilityRole="button"
          accessibilityLabel={post.liked ? 'Unlike post' : 'Like post'}
          accessibilityState={{ selected: post.liked }}
          onPress={onLike}
          style={({ pressed }) => [
            styles.actionButton,
            post.liked && styles.activeActionButton,
            pressed && styles.pressedActionButton,
          ]}
        >
          <Image
            source={post.liked ? actionIcons.liked : actionIcons.like}
            accessible={false}
            style={[styles.actionIcon, post.liked && styles.activeActionIcon]}
          />
          <Text style={[styles.actionText, post.liked && styles.activeActionText]}>
            {post.liked ? 'Liked' : 'Likes'}
          </Text>
```

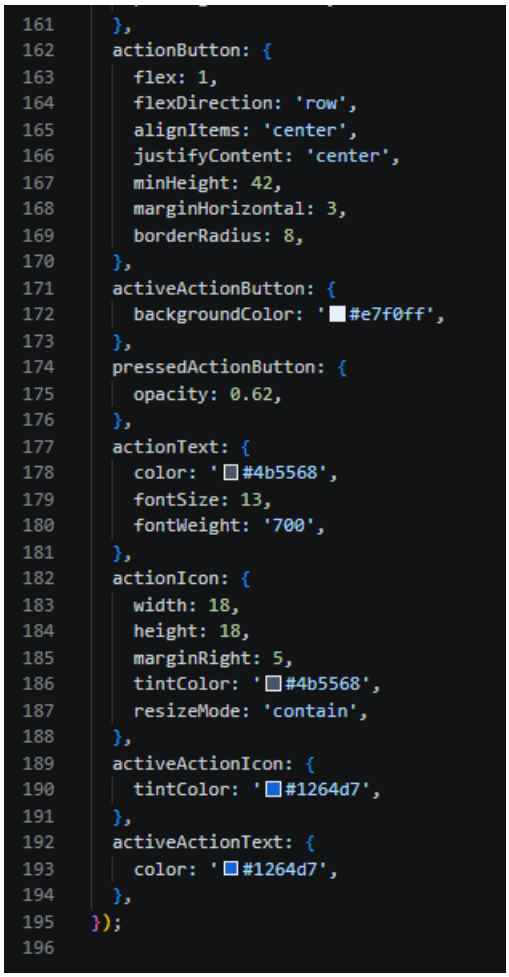
```
        </Pressable>

        <Pressable
          accessibilityRole="button"
          accessibilityLabel="Add a comment"
          onPress={onComment}
          style={({ pressed }) => [styles.actionButton, pressed && styles.pressedActionButton]}
        >
          <Image source={actionIcons.comment} accessible={false} style={styles.actionIcon} />
          <Text style={styles.actionText}>Comments</Text>
        </Pressable>

        <Pressable
          accessibilityRole="button"
          accessibilityLabel="Share post"
          onPress={onShare}
          style={({ pressed }) => [styles.actionButton, pressed && styles.pressedActionButton]}
        >
          <Image source={actionIcons.share} accessible={false} style={styles.actionIcon} />
          <Text style={styles.actionText}>Shares</Text>
        </Pressable>
      </View>
    </View>
  );
}

const styles = StyleSheet.create({
  card: {
    marginBottom: 14,
    backgroundColor: '#ffffff',
    borderRadius: 14,
    shadowColor: '#14213d',
    shadowOffset: {
      width: 0,
      height: 3,
    },
    shadowOpacity: 0.1,
    shadowRadius: 8,
    elevation: 3,
    overflow: 'hidden',
  },
  authorRow: {
    flexDirection: 'row',
    alignItems: 'center',
    paddingHorizontal: 14,
    paddingTop: 14,
  },
  avatar: {
    width: 46,
    height: 46,
    borderRadius: 23,
    backgroundColor: '#dfe7f3',
  },
```

```
  authorText: {
    flex: 1,
    marginLeft: 11,
  },
  username: {
    color: '#172033',
    fontSize: 16,
    fontWeight: '700',
  },
  postTime: {
    marginTop: 2,
    color: '#778198',
    fontSize: 12,
  },
  content: {
    paddingHorizontal: 14,
    paddingVertical: 12,
    color: '#30394c',
    fontSize: 15,
    lineHeight: 21,
  },
  postImageFrame: {
    marginHorizontal: 14,
    borderRadius: 10,
    overflow: 'hidden',
    backgroundColor: '#dfe7f3',
  },
  postImage: {
    width: '100%',
    height: 230,
    borderRadius: 10,
    resizeMode: 'cover',
  },
  statisticsRow: {
    flexDirection: 'row',
    justifyContent: 'space-between',
    paddingHorizontal: 14,
    paddingVertical: 11,
  },
  statisticsText: {
    color: '#667085',
    fontSize: 12,
    fontWeight: '600',
  },
  divider: {
    height: StyleSheet.hairlineWidth,
    marginHorizontal: 14,
    backgroundColor: '#ccd3df',
  },
  actionsRow: {
    flexDirection: 'row',
    paddingHorizontal: 8,
    paddingVertical: 7,
```



```
  },
  actionButton: {
    flex: 1,
    flexDirection: 'row',
    alignItems: 'center',
    justifyContent: 'center',
    minHeight: 42,
    marginHorizontal: 3,
    borderRadius: 8,
  },
  activeActionButton: {
    backgroundColor: '#e7f0ff',
  },
  pressedActionButton: {
    opacity: 0.62,
  },
  actionText: {
    color: '#4b5568',
    fontSize: 13,
    fontWeight: '700',
  },
  actionIcon: {
    width: 18,
    height: 18,
    marginRight: 5,
    tintColor: '#4b5568',
    resizeMode: 'contain',
  },
  activeActionIcon: {
    tintColor: '#1264d7',
  },
  activeActionText: {
    color: '#1264d7',
  },
});

```

posts.js

Bai1 > src > data > JS posts.js > ...

```js
// 23521720 - Nguyen Minh Tuan
const initialPosts = [
  {
    id: '1',
    username: 'Linh Nguyen',
    avatar: 'https://i.pravatar.cc/160?img=47',
    content: 'A quiet morning by the lake is the best way to reset for a new week.',
    image:
      'https://images.unsplash.com/photo-1501785888041-af3ef285b470?auto=format&fit=crop&w=1200&q=85',
    likes: 128,
    comments: 24,
    shares: 9,
    liked: false,
  },
  {
    id: '2',
    username: 'Minh Anh',
    avatar: 'https://i.pravatar.cc/160?img=32',
    content: 'Found a colorful corner of the city today. Every street has a story.',
    image:
      'https://images.unsplash.com/photo-1477959858617-67f85cf4f1df?auto=format&fit=crop&w=1200&q=85',
    likes: 256,
    comments: 41,
    shares: 18,
    liked: false,
  },
  {
    id: '3',
    username: 'Quang Huy',
    avatar: 'https://i.pravatar.cc/160?img=12',
    content: 'Nothing beats the sound of waves and a sky full of warm sunset colors.',
    image:
      'https://images.unsplash.com/photo-1500530855697-b586d89ba3ee?auto=format&fit=crop&w=1200&q=85',
    likes: 392,
    comments: 67,
    shares: 31,
    liked: false,
  },
  {
    id: '4',
    username: 'Bao Tran',
    avatar: 'https://i.pravatar.cc/160?img=5',
    content: 'Weekend hike completed. Tired legs, clear mind, and a wonderful view.',
    image:
      'https://images.unsplash.com/photo-1551632811-561732d1e306?auto=format&fit=crop&w=1200&q=85',
    likes: 175,
    comments: 29,
    shares: 12,
    liked: false,
  },
];

export default initialPosts;

```

Màn hình kết quả



Social Media Feed

Connect, share, and enjoy



392 Likes 67 Comments 31 Shares

Likes Comments Shares



Bao Tran

Just now

Weekend hike completed. Tired legs, clear mind, and a wonderful view.



175 Likes 32 Comments 15 Shares

Likes Comments Shares

Social Media Feed

Connect, share, and enjoy



392 Likes 67 Comments 31 Shares

Likes Comments Shares



Bao Tran

Just now

Weekend hike completed. Tired legs, clear mind, and a wonderful view.



176 Likes 34 Comments 17 Shares

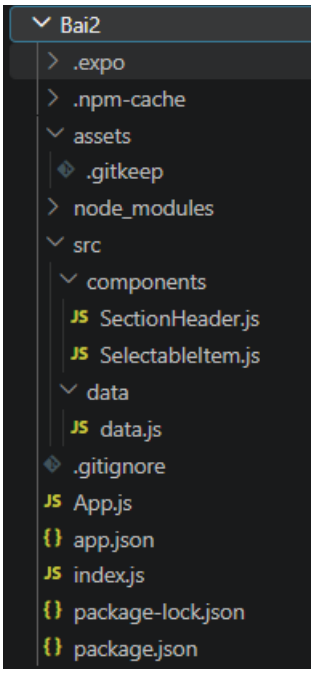
Liked Comments Shares

Bài 2: Workouts, Fruits and Vegetables

Link video

https://drive.google.com/file/d/1oyk3yvJKpvFTi0RALwXEcJ7s4KYK-4bZ/view?usp=sharing

Cấu trúc project



Các file code

App.js

Bai2 > JS App.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { useState } from 'react';
import {
  FlatList,
  ImageBackground,
  Platform,
  SafeAreaView,
  ScrollView,
  SectionList,
  StatusBar as NativeStatusBar,
  StyleSheet,
  Text,
  View,
} from 'react-native';
import { StatusBar } from 'expo-status-bar';

import SectionHeader from './src/components/SectionHeader';
import SelectableItem from './src/components/SelectableItem';
import { backgroundImages, fruitsVegetables, workouts } from './src/data/data';

export default function App() {
  const [selectedItems, setSelectedItems] = useState([]);

  const toggleSelection = (itemName) => {
    setSelectedItems((currentItems) =>
      currentItems.includes(itemName)
        ? currentItems.filter((item) => item !== itemName)
        : [...currentItems, itemName],
    );
  };

  const renderWorkout = ({ item }) => (
    <SelectableItem
      label={item.type}
      selected={selectedItems.includes(item.type)}
      onPress={() => toggleSelection(item.type)}
    />
  );

  const renderProduce = ({ item }) => (
    <SelectableItem
      label={item}
      selected={selectedItems.includes(item)}
      onPress={() => toggleSelection(item)}
    />
  );

  const renderSectionHeader = ({ section }) => (
    <SectionHeader title={section.title} imageUrl={section.url} />
  );

  return (
    <SafeAreaView style={styles.safeArea}>
      <StatusBar style="light" />
```

```
      <View style={styles.appHeader}>
        <Text style={styles.appTitle}>Healthy Choices</Text>
        <Text style={styles.appSubtitle}>Build your workout and food list</Text>
      </View>

      <View style={styles.listsArea}>
        <View style={styles.listPanel}>
          <Text style={styles.listTitle}>FlatList - Workouts</Text>
          <ImageBackground
            source={{ uri: backgroundImages.workouts }}
            style={styles.background}
            imageStyle={styles.backgroundImage}
          >
            <FlatList
              data={workouts}
              extraData={selectedItems}
              renderItem={renderWorkout}
              keyExtractor={(item) => item.id}
              contentContainerStyle={styles.listContent}
              showsVerticalScrollIndicator={false}
            />
          </ImageBackground>
        </View>

        <View style={styles.listPanel}>
          <Text style={styles.listTitle}>SectionList - Fruits & Vegetables</Text>
          <ImageBackground
            source={{ uri: backgroundImages.produce }}
            style={styles.background}
            imageStyle={styles.backgroundImage}
          >
            <SectionList
              sections={fruitsVegetables}
              extraData={selectedItems}
              renderItem={renderProduce}
              renderSectionHeader={renderSectionHeader}
              keyExtractor={(item, index) => `${item}-${index}`}
              contentContainerStyle={styles.listContent}
              stickySectionHeadersEnabled={false}
              showsVerticalScrollIndicator={false}
            />
          </ImageBackground>
        </View>
      </View>

      <View style={styles.selectedPanel}>
        <Text style={styles.selectedTitle}>SELECTED ITEMS:</Text>
        <ScrollView style={styles.selectedScroll} showsVerticalScrollIndicator={false}>
          <Text style={styles.selectedText}>
            {selectedItems.length > 0 ? selectedItems.join(', ') : 'None'}
          </Text>
        </ScrollView>
      </View>
```

```
    </SafeAreaView>
  );
}

const styles = StyleSheet.create({
  safeArea: {
    flex: 1,
    backgroundColor: '#edf2f7',
  },
  appHeader: {
    paddingHorizontal: 18,
    paddingTop: Platform.OS === 'android' ? NativeStatusBar.currentHeight + 11 : 11,
    paddingBottom: 12,
    backgroundColor: '#173f6b',
  },
  appTitle: {
    color: '#ffffff',
    fontSize: 22,
    fontWeight: '800',
  },
  appSubtitle: {
    marginTop: 2,
    color: '#cfe3f7',
    fontSize: 12,
  },
  listsArea: {
    flex: 1,
    paddingHorizontal: 10,
    paddingTop: 8,
  },
  listPanel: {
    flex: 1,
    minHeight: 0,
    marginBottom: 8,
    borderWidth: 1,
    borderColor: '#cbd7e6',
    borderRadius: 12,
    backgroundColor: '#ffffff',
    overflow: 'hidden',
  },
  listTitle: {
    paddingHorizontal: 13,
    paddingVertical: 9,
    color: '#173f6b',
    fontSize: 16,
    fontWeight: '800',
    textAlign: 'center',
    backgroundColor: '#ffffff',
  },
  background: {
    flex: 1,
    backgroundColor: '#dfe8f2',
  },
```

```
  backgroundImage: {
    opacity: 0.36,
  },
  listContent: {
    paddingVertical: 5,
  },
  selectedPanel: {
    maxHeight: 112,
    marginHorizontal: 10,
    marginBottom: 10,
    paddingHorizontal: 13,
    paddingTop: 10,
    paddingBottom: 8,
    borderWidth: 1,
    borderColor: '#f0bd5b',
    borderRadius: 12,
    backgroundColor: '#fff7e6',
  },
  selectedTitle: {
    color: '#a65508',
    fontSize: 13,
    fontWeight: '900',
    letterSpacing: 0.4,
  },
  selectedScroll: {
    marginTop: 4,
  },
  selectedText: {
    color: '#4f3b22',
    fontSize: 13,
    lineHeight: 18,
  },
});

```

SelectableItem.js

Bai2 > src > components > JS SelectableItem.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { Pressable, StyleSheet, Text, View } from 'react-native';

export default function SelectableItem({ label, selected, onPress }) {
  return (
    <View style={[styles.item, selected && styles.selectedItem]}>
      <Text style={[styles.label, selected && styles.selectedLabel]}>{label}</Text>
      <Pressable
        accessibilityRole="button"
        accessibilityLabel={`${selected ? 'Deselect' : 'Select'} ${label}`}
        onPress={onPress}
        style={({ pressed }) => [
          styles.button,
          selected && styles.deselectButton,
          pressed && styles.pressedButton,
        ]}
      >
        <Text style={styles.buttonText}>{selected ? 'DESELECT' : 'SELECT'}</Text>
      </Pressable>
    </View>
  );
}

const styles = StyleSheet.create({
  item: {
    flexDirection: 'row',
    alignItems: 'center',
    minHeight: 48,
    marginHorizontal: 10,
    marginVertical: 4,
    paddingLeft: 13,
    paddingRight: 7,
    paddingVertical: 6,
    borderWidth: 1,
    borderColor: '#d8e0ec',
    borderRadius: 10,
    backgroundColor: 'rgba(255, 255, 255, 0.93)',
  },
  selectedItem: {
    borderColor: '#16886f',
    backgroundColor: 'rgba(226, 250, 243, 0.96)',
  },
  label: {
    flex: 1,
    marginRight: 8,
    color: '#243047',
    fontSize: 14,
    fontWeight: '600',
  },
  selectedLabel: {
    color: '#0c604f',
  },
  button: {
    minWidth: 82,
```



```
    alignItems: 'center',
    paddingHorizontal: 10,
    paddingVertical: 8,
    borderRadius: 8,
    backgroundColor: '#1769d2',
  },
  deselectButton: {
    backgroundColor: '#d94a4a',
  },
  pressedButton: {
    opacity: 0.68,
  },
  buttonText: {
    color: '#ffffff',
    fontSize: 11,
    fontWeight: '800',
    letterSpacing: 0.3,
  },
});

```

SectionHeader.js

Bai2 > src > components > JS SectionHeader.js > ...

```
// 23521720 - Nguyen Minh Tuan
import { Image, StyleSheet, Text, View } from 'react-native';

export default function SectionHeader({ title, imageUrl }) {
  return (
    <View style={styles.header}>
      <Image source={{ uri: imageUrl }} style={styles.icon} />
      <Text style={styles.title}>{title}</Text>
    </View>
  );
}

const styles = StyleSheet.create({
  header: {
    flexDirection: 'row',
    alignItems: 'center',
    marginHorizontal: 10,
    marginTop: 7,
    marginBottom: 3,
    paddingHorizontal: 12,
    paddingVertical: 8,
    borderRadius: 10,
    backgroundColor: 'rgba(24, 50, 84, 0.92)',
  },
  icon: {
    width: 30,
    height: 30,
    marginRight: 10,
  },
  title: {
    color: '#ffffff',
    fontSize: 16,
    fontWeight: '800',
  },
});

```

data.js

Bai2 > src > data > JS data.js > ...

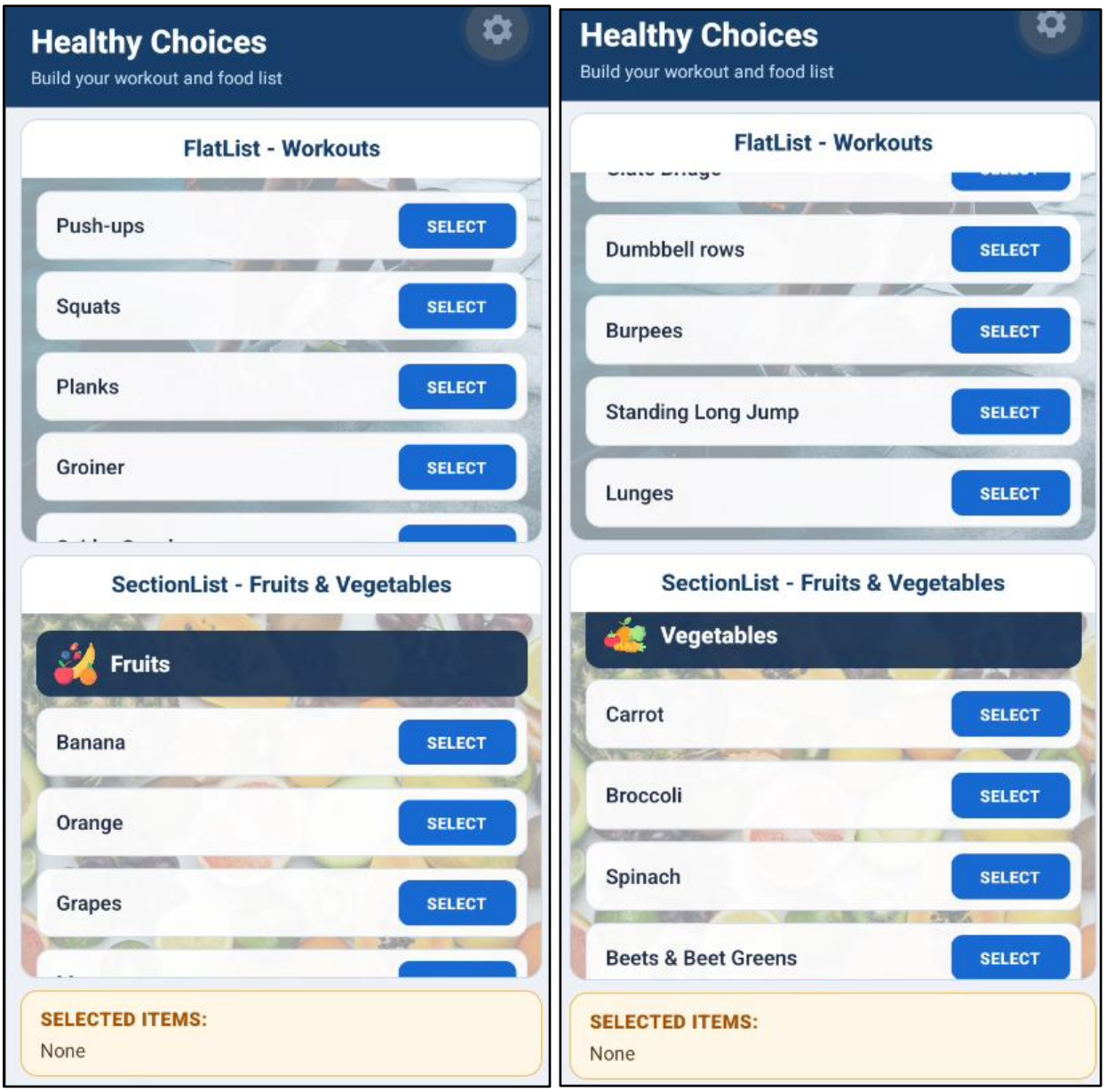
```
// 23521720 - Nguyen Minh Tuan
export const workouts = [
  { id: '1', type: 'Push-ups' },
  { id: '2', type: 'Squats' },
  { id: '3', type: 'Planks' },
  { id: '4', type: 'Groiner' },
  { id: '5', type: 'Spider Crawl' },
  { id: '6', type: 'Glute Bridge' },
  { id: '7', type: 'Dumbbell rows' },
  { id: '8', type: 'Burpees' },
  { id: '9', type: 'Standing Long Jump' },
  { id: '10', type: 'Lunges' },
];

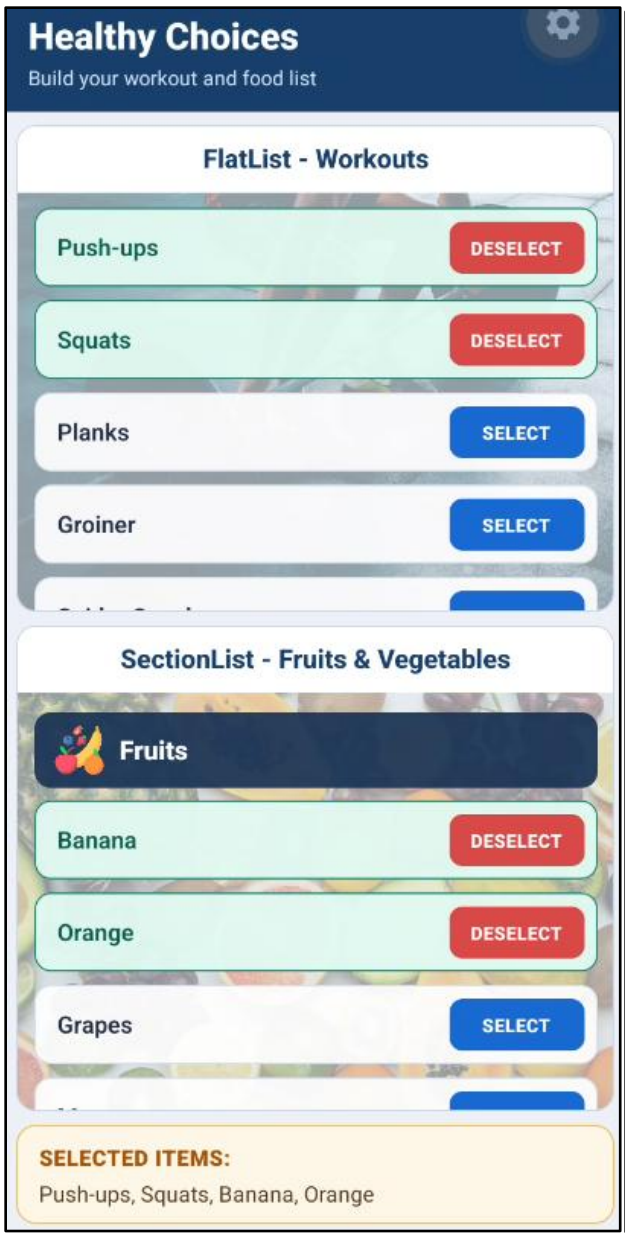
export const fruitsVegetables = [
  {
    title: 'Fruits',
    url: 'https://cdn-icons-png.flaticon.com/512/1625/1625099.png',
    data: ['Banana', 'Orange', 'Grapes', 'Mango', 'Pineapple'],
  },
  {
    title: 'Vegetables',
    url: 'https://cdn-icons-png.flaticon.com/512/2153/2153788.png',
    data: ['Carrot', 'Broccoli', 'Spinach', 'Beets & Beet Greens', 'Kale'],
  },
];

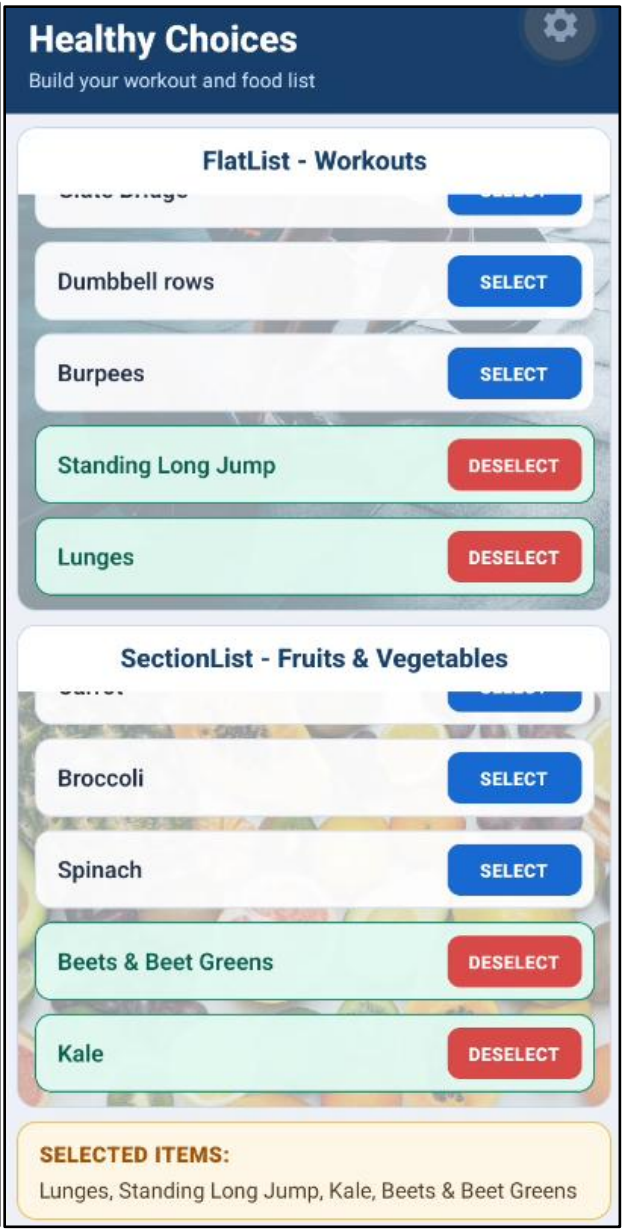
export const backgroundImages = {
  workouts:
    'https://images.unsplash.com/photo-1517836357463-d25dfeac3438?auto=format&fit=crop&w=1200&q=80',
  produce:
    'https://images.unsplash.com/photo-1610832958506-aa56368176cf?auto=format&fit=crop&w=1200&q=80',
};

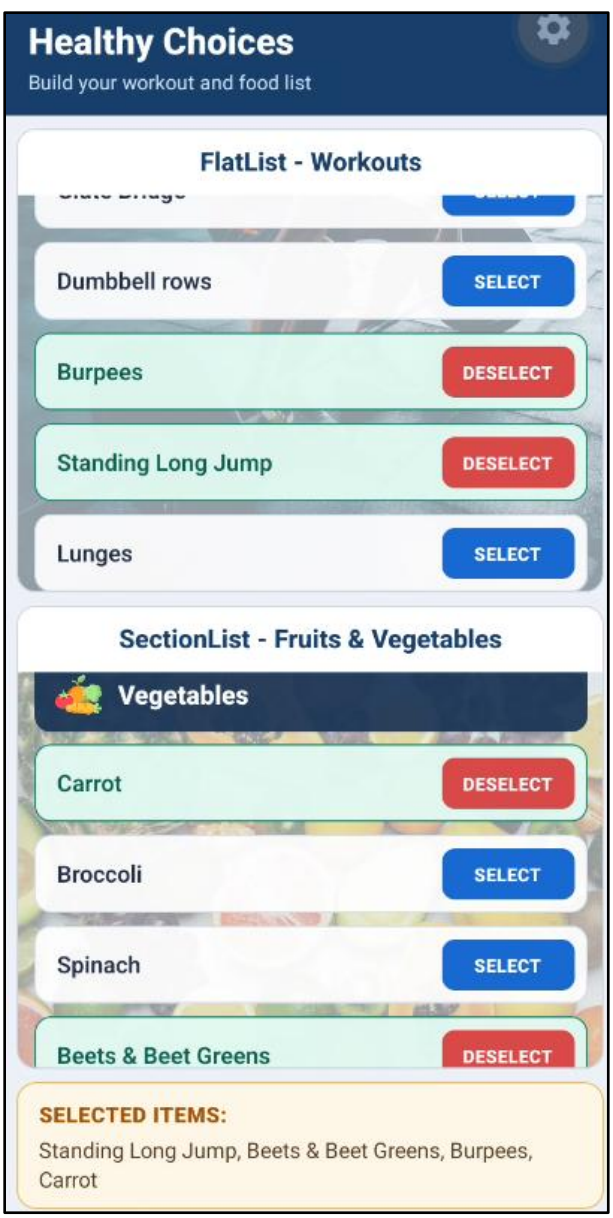
```

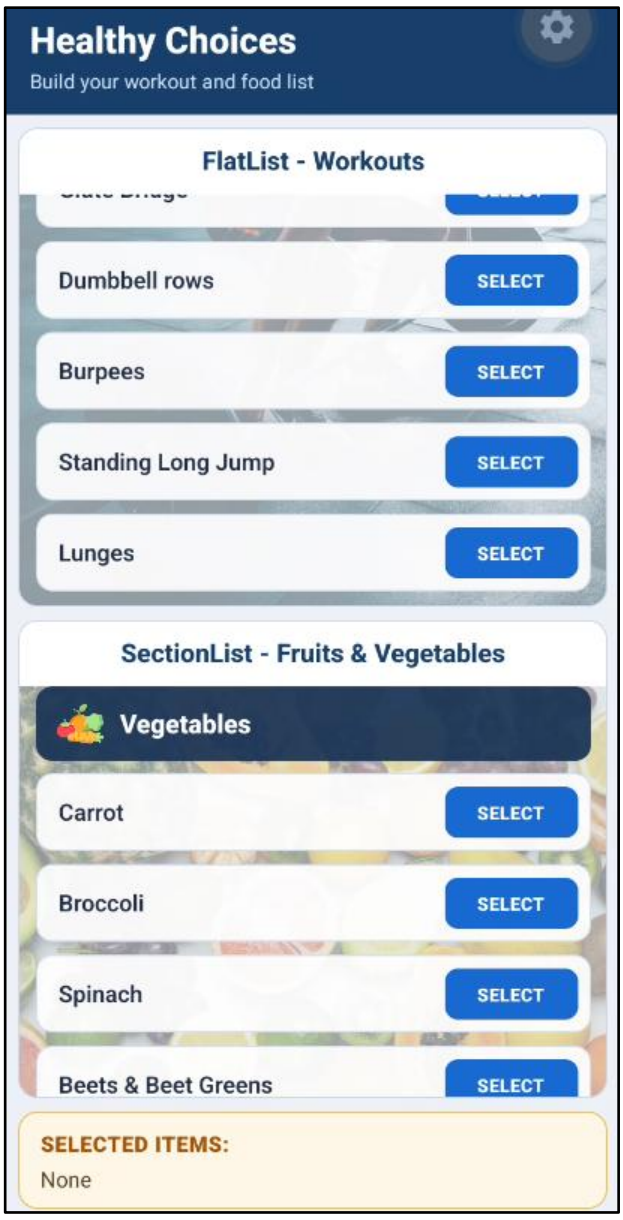
Màn hình kết quả




Healthy Choices
Build your workout and food list
FlatList - Workouts
Push-ups
DESELECT
Squats
DESELECT
Planks
SELECT
Groiner
SELECT
SectionList - Fruits & Vegetables
Fruits
Banana
DESELECT
Orange
DESELECT
Grapes
SELECT
SELECTED ITEMS:
Push-ups, Squats, Banana, Orange


Healthy Choices
Build your workout and food list
FlatList - Workouts
Dumbbell rows
SELECT
Burpees
SELECT
Standing Long Jump
DESELECT
Lunges
DESELECT
SectionList - Fruits & Vegetables
Broccoli
SELECT
Spinach
SELECT
Beets & Beet Greens
DESELECT
Kale
DESELECT
SELECTED ITEMS:
Lunges, Standing Long Jump, Kale, Beets & Beet Greens


Healthy Choices
Build your workout and food list
FlatList - Workouts
Dumbbell rows
SELECT
Burpees
DESELECT
Standing Long Jump
DESELECT
Lunges
SELECT
SectionList - Fruits & Vegetables
Vegetables
Carrot
DESELECT
Broccoli
SELECT
Spinach
SELECT
Beets & Beet Greens
DESELECT
SELECTED ITEMS:
Standing Long Jump, Beets & Beet Greens, Burpees, Carrot


Healthy Choices
Build your workout and food list
FlatList - Workouts
Dumbbell rows
SELECT
Burpees
SELECT
Standing Long Jump
SELECT
Lunges
SELECT
SectionList - Fruits & Vegetables
Vegetables
Carrot
SELECT
Broccoli
SELECT
Spinach
SELECT
Beets & Beet Greens
SELECT
SELECTED ITEMS:
None